

Question Paper

Exam Date & Time: 19-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
SUBJECT: ECE 3121/ECE_3121 ANALOG AND DIGITAL COMMUNICATION
(SPL: ELECTRONICS & COMMUNICATION ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

Missing data may be suitably assumed.

- 1A) With relevant mathematical expressions and block diagram explain the working of switching modulator for AM generation. (5)
Mention the specification of filter to be used to obtain the desired AM signal.

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1B) Write down the expression for NBFM signal. Given a maximum frequency deviation 1 kHz and a highest modulating frequency 5 kHz, with $\beta \ll 1$. Compute the modulation index and the FM bandwidth for this condition. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Determine the Fourier Transform of the signal $x(t) = \cos(6\pi f_c t)$. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) Consider three signals $s_1(t), s_2(t), s_3(t)$ defined over $0 \leq t \leq T$: (5)

$$s_1(t) = \begin{cases} 2, & 0 \leq t \leq T \\ 0, & \text{elsewhere} \end{cases}$$
$$s_2(t) = \begin{cases} 4, & 0 \leq t \leq T/2 \\ 0, & \text{elsewhere} \end{cases}$$
$$s_3(t) = \begin{cases} -2, & 0 \leq t \leq T \\ 0, & \text{elsewhere} \end{cases}$$

Determine orthonormal basis functions using Gram-Schmidt and express the signals in terms of these basis functions. Sketch the resulting orthonormal functions.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2B) Illustrate how the matched filter impulse response is obtained as a time-reversed and delayed version of the input signal. Apply this property to explain why the output is maximized at $t = T$. (3)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 2C) (2)

Assume that three signals $s_1(t), s_2(t)$, and $s_3(t)$ can be represented using two orthonormal basis functions $\Phi_1(t)$ (assume in x -axis) and $\Phi_2(t)$ (assume in y -axis).

The coordinates of these signals are

$$s_1 = (5, 1), s_2 = (-1, -1), s_3 = (-1, -2).$$

Draw the constellation diagram and express the three signals $s_1(t), s_2(t)$, and $s_3(t)$ as a linear combination of the given basis functions.

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 3A) With the help of a neat block diagram, describe the duo-binary conversion system. Obtain the frequency response of duo-binary encoder and hence derive the impulse response. Also plot the magnitude and phase responses of the system. (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3B) Consider the 4-bit binary sequence "1100". Obtain the discrete PAM representation of the sequence using RZ unipolar format. Calculate the autocorrelation function of the RZ unipolar format assuming that the input data is statistically independent and equally likely. (3)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- 3C) Sketch the typical eye pattern and describe Inter-symbol Interference performance parameters. (2)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- 4A) Derive the expression for average probability of error at the receiver for a binary-encoded PCM wav that uses nonreturn-to-zero unipolar (ON-OFF) format, assuming AWGN channel. Plot the corresponding signal space diagram with decision boundary and likelihood functions. (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 4B) Consider transmission of a dibit 00 using QPSK modulation, where the conditional probability of joint event is $x_1 < 0$ and $x_2 < 0$. Determine average probability of symbol error with all intermediate steps. Assume AWGN channel. (3)
- Course Outcome: "CO4" / Cognitive Level: "Application"
- 4C) The binary data sequence 1 0 1 1 0 0 1 0 0 is applied to the input of a DPSK transmitter. Determine the differentially encoded sequence and transmitted phase. Assume initial reference bit as 1. (2)
- Course Outcome: "CO4" / Cognitive Level: "Application"
- 5A) Construct minimum variance Huffman code for the given source probabilities $p_k=[0.12, 0.11, 0.41, 0.19, 0.17]$. Also determine the average code length of this Huffman code. (5)
- Course Outcome: "CO5" / Cognitive Level: "Application"
- 5B) Determine the entropy, efficiency and redundancy of the Huffman code as given in Q5A. (3)
- Course Outcome: "CO5" / Cognitive Level: "Application"
- 5C) Determine the minimum variance of this Huffman code as given in Q5A. (2)
- Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_3121:ANALOG AND DIGITAL COMMUNICATION

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	15
3	3	CO 3	10
4	4	CO 4	5
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_3121:ANALOG AND DIGITAL COMMUNICATION

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	15(30%)
3	Application	10	35(70%)
4	Analysis	0	0(0%)

5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)